NeonPine – MCU netlist for firmware

Generated by `tools/mcu_netlist.py` from `NeonPine.kicad_sch` – do not edit by hand, rerun the script after pin changes.

MCU: **U1 CH32V003F4P6** (TSSOP-20), **VDD = 5 V**, HSI ×2 = 48 MHz (needed by rv003usb).

U1 pins

Pin	Port	Net	Firmware use	Connects to
1	PD4	USB_DM_MCU	USB D− (rv003usb `USB_PIN_DM 4`; EXTI4 falling edge, highest IRQ priority)	R14 68 → **USB_DM**: D5.1 (BZX884-C3V6), J1.A7, J1.B7, R15.2 (1k5)
2	PD5	SR_LATCH	74HC595 RCLK (U2, U3): rising edge copies the shift registers to the outputs	U2.12; U3.12
3	PD6	GATE_DRV	Boost MOSFET gate (via R1 47R, R2 100k pull-down): **TIM2_CH3, AFIO TIM2_RM = 11** (full remap), ≈30 kHz, ≤60 % duty	R1 47 → **GATE**: Q1.1, R2.1 (100k)
4	PD7	SR_DATA	74HC595 SER of U2. **PD7 = NRST pin: configure NRST as GPIO** (option byte RST_MODE, `minichlink -D`)	U2.14
5	PA1	LDR	Light sensor TEPT4400 (emitter, 22k to GND), **ADC A1**	Q2.2; R9.1 (22k)
6	PA2	TOUCH2	Trunk touch pad TP2 via R11 1k, **ADC A0** (TKEY)	R11 1k → **Net-(R11-Pad1)**: TP2.1
7	VSS	GND	VSS	ground pours
8	PD0	LAMP8	LA8 K side: lamp pad 1 switch (Q18 unit A via R48 22k); high = on = that electrode glows	R48 22k → **Net-(Q18A-B1)**: Q18.2
9	VDD	+5V	VDD (USB 5 V after PTC + ferrite)	C6 100n at the pin; supply rail
10	PC0	LAMP7	LA7 K side: lamp pad 1 switch (Q17 unit A via R47 22k); high = on = that electrode glows	R47 22k → **Net-(Q17A-B1)**: Q17.2
11	PC1	LAMP4	LA4 K side: lamp pad 1 switch (Q14 unit A via R44 22k); high = on = that electrode glows	R44 22k → **Net-(Q14A-B1)**: Q14.2
12	PC2	LAMP2	LA2 K side: lamp pad 1 switch (Q12 unit A via R42 22k); high = on = that electrode glows	R42 22k → **Net-(Q12A-B1)**: Q12.2
13	PC3	LAMP5	LA5 K side: lamp pad 1 switch (Q15 unit A via R45 22k); high = on = that electrode glows	R45 22k → **Net-(Q15A-B1)**: Q15.2
14	PC4	LAMP3	LA3 K side: lamp pad 1 switch (Q13 unit A via R43 22k); high = on = that electrode glows	R43 22k → **Net-(Q13A-B1)**: Q13.2
15	PC5	LAMP9	LA9 K side: lamp pad 1 switch (Q19 unit A via R49 22k); high = on = that electrode glows	R49 22k → **Net-(Q19A-B1)**: Q19.2
16	PC6	LAMP6	LA6 K side: lamp pad 1 switch (Q16 unit A via R46 22k); high = on = that electrode glows	R46 22k → **Net-(Q16A-B1)**: Q16.2
17	PC7	SR_CLK	74HC595 SRCLK (U2, U3), rising edge shifts; also test pad TP3	TP3.1; U2.11; U3.11
18	PD1	SWIO	SWD programming / debug (pogo pads J2: 5V, SWIO, GND)	J2.2
19	PD2	USB_DP_MCU	USB D+ (rv003usb `USB_PIN_DP 2`)	R13 68 → **USB_DP**: D4.1 (BZX884-C3V6), J1.A6, J1.B6
20	PD3	VFB	HV feedback, **ADC A4**: V_HV · 39k / 2.039M (105 V → 2.0 V), 10n filter	C5.1 (10n); R6 1M → **Net-(R5-Pad2)**: R5.2 (1M); R7.1 (39k)

74HC595 chain (U2 → U3)

`SR_DATA` → U2 SER, U2 QH′ → U3 SER (`SR_CHAIN`); clock and latch are shared. /OE is tied low and /SRCLR high, so the outputs are random until the first latch (the boost is held off by R2 meanwhile). Shift a 16-bit word **MSB first** (`SR_DATA` valid before the `SR_CLK` rising edge), then pulse `SR_LATCH`. Word bit *b* ends up on:

Word bit	Output	Pin	Net	Drives
15	U3 QH	7	LAMPA6	LA6 A side: lamp pad 2 switch (Q16 unit B via R56 22k); high = on = that electrode glows
14	U3 QG	6	–	– (not connected)
13	U3 QF	5	LAMPA9	LA9 A side: lamp pad 2 switch (Q19 unit B via R59 22k); high = on = that electrode glows
12	U3 QE	4	LAMP10	LA10 K side: lamp pad 1 switch (Q20 unit A via R50 22k); high = on = that electrode glows
11	U3 QD	3	LAMPA10	LA10 A side: lamp pad 2 switch (Q20 unit B via R60 22k); high = on = that electrode glows
10	U3 QC	2	–	– (not connected)
9	U3 QB	1	–	– (not connected)
8	U3 QA	15	–	– (not connected)
7	U2 QH	7	LAMPA3	LA3 A side: lamp pad 2 switch (Q13 unit B via R53 22k); high = on = that electrode glows
6	U2 QG	6	LAMPA5	LA5 A side: lamp pad 2 switch (Q15 unit B via R55 22k); high = on = that electrode glows
5	U2 QF	5	LAMP1	LA1 K side: lamp pad 1 switch (Q11 unit A via R41 22k); high = on = that electrode glows
4	U2 QE	4	LAMPA1	LA1 A side: lamp pad 2 switch (Q11 unit B via R51 22k); high = on = that electrode glows
3	U2 QD	3	LAMPA2	LA2 A side: lamp pad 2 switch (Q12 unit B via R52 22k); high = on = that electrode glows
2	U2 QC	2	LAMPA4	LA4 A side: lamp pad 2 switch (Q14 unit B via R54 22k); high = on = that electrode glows
1	U2 QB	1	LAMPA7	LA7 A side: lamp pad 2 switch (Q17 unit B via R57 22k); high = on = that electrode glows
0	U2 QA	15	LAMPA8	LA8 A side: lamp pad 2 switch (Q18 unit B via R58 22k); high = on = that electrode glows

Lamps

Each lamp sits in an H-bridge (DESIGN.md §2.3): turning the **K-side** switch on makes lamp pad 1 the cathode (that electrode glows), the **A-side** switch makes pad 2 the cathode. Never turn both on (dark, but both 150k pull-ups draw current).

Lamp	K side (net LAMPn)	A side (net LAMPAn)
LA1	U2 QF = **SR bit 5**	U2 QE = **SR bit 4**
LA2	U1 pin 12 = **PC2**	U2 QD = **SR bit 3**
LA3	U1 pin 14 = **PC4**	U2 QH = **SR bit 7**
LA4	U1 pin 11 = **PC1**	U2 QC = **SR bit 2**
LA5	U1 pin 13 = **PC3**	U2 QG = **SR bit 6**
LA6	U1 pin 16 = **PC6**	U3 QH = **SR bit 15**
LA7	U1 pin 10 = **PC0**	U2 QB = **SR bit 1**
LA8	U1 pin 8 = **PD0**	U2 QA = **SR bit 0**

Lamp	K side (net LAMPn)	A side (net LAMPAn)
LA9	U1 pin 15 = **PC5**	U3 QF = **SR bit 13**
LA10	U3 QE = **SR bit 12**	U3 QD = **SR bit 11**

Peripheral / option-byte summary

- **Option bytes:** NRST disabled (RST_MODE = GPIO) – PD7 carries SR_DATA. Keep it disabled if the rv003usb bootloader is configured (its `configurebootloader` sets OB_RST_EN_DT1ms by default).
- **AFIO:** TIM2_RM = 11 (TIM2_CH3 → PD6). The other TIM2 channels then map to PC1/PC7/PD5 – leave their outputs disabled.
- **Timers:** TIM2 = boost PWM (CH3). TIM1 and SysTick are free (BAM frame, UI timing).
- **ADC:** A0 = PA2 touch, A1 = PA1 light sensor, A4 = PD3 HV feedback (calibrate against Vrefint).
- **USB (rv003usb `usb_config.h`):** `USB_PORT D`, `USB_PIN_DP 2`, `USB_PIN_DM 4`, no `USB_PIN_DPU` (R15 1k5 pulls D− to VBUS permanently; force re-enumeration by driving D− low briefly). EXTI line 4 is taken; its handler must be the highest priority and never be preempted – keep other ISRs short or preemptible.
- **Reset state:** all pins are floating inputs. R2 holds the boost gate low, so HV+ stays at ≈ 5 V and the lamp switches (floating bases, no pull-downs) cannot light anything; drive the lamp pins and latch the 595s before starting TIM2.